

Proteins and amino acids

- What are Functions of Proteins?
- What Are Amino Acids?
- What Are Zwitterions?
- What Determines the Characteristics of Amino Acids?
- What Are Uncommon Amino Acids?
- How Do Amino Acids Combine to Form Proteins?
- What Are the Properties of Proteins?
- What are the Structures of a Protein?
- How Are Proteins Denatured?

What are proteins?

Proteins are polypeptides with more than 50 amino acid residues with a molecular weight greater than 5000. They may attach to variety of *cofactors*, carbohydrates and ions.

These *macromolecules* show great diversity in their physical properties, ranging from water-soluble enzymes to the insoluble keratin of hair, which is reflected in a wide range of biological functions.

They are most abundant intracellular macromolecules and constitute over half the dry weight of most organisms.

All proteins in human are polymers of (20) different monomer or amino acids, and they arranged in a specific order that determines the characteristics of the protein and its function in the cell.

Proteins like carbohydrate and lipids composed of C, H, and O, while proteins also contain Nitrogen.

The nitrogen content is around 16% of the molecular weight of proteins. Small amounts of S and P are also present.

Importance of protein:-

- Proteins are the main structural components of the cytoskeleton. They are the source to replace Nitrogen of the body.
- Biochemical catalysts known as enzymes are proteins.
- Proteins known as immunoglobulin's serve as the first line of defense against bacterial and viral infections.
- Several hormones are protein in nature such as insulin.

- Structural proteins provide mechanical support and some of them like actin and myosin are contractile proteins and help in the movement of muscle fiber.
- Some proteins present in cell membrane, cytoplasm and nucleus of the cell act as receptors.
- Transport proteins carry specific substances either across the membrane or in the body fluids.
- Storage proteins bind with specific substances and store them, e.g. iron is stored as ferritin.

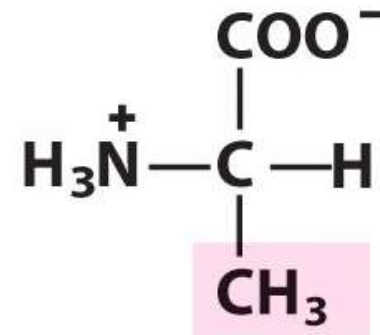
Amino acids

As many as 300 amino acids occur in nature. Of these, only 20 known as (proteogenic) standard amino acids.

Amino acids are the simplest units of a protein molecule and they form the building blocks of protein structure.

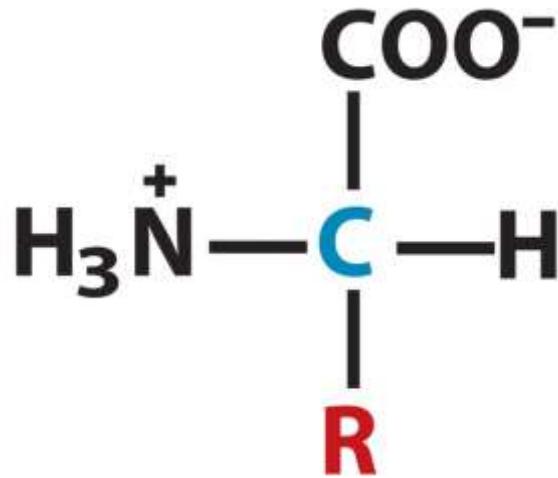
An amino acid is a molecule that contains an *amino* group and a *carboxyl* group. The amino and carboxyl groups are *both* attached to the α -carbon atom

At neutral pH they exist as dipolar ions commonly referred to as zwitterions (from the German *meaning* “hybrid.”), due to the negatively charged carboxyl group and a positively charged amino group.



Alanine

The unique characteristics of the (20) amino acids are due to a side chain (R) group, which can be an alkyl, hydroxyl, aromatic, or heterocyclic group.



General structure of (α) amino acids.

Although there are many different types of amino acid (for example, nonprotein amino acids used in metabolism), only a subset of these exist in proteins. These are the α -*amino acids*.

All the amino acids have trivial or common names based on the source from which they were first isolated or based on their properties. For example:

Asparagine was named so, as it was isolated from asparagus and glycine named because of its sweet taste (Greek: 'glykos' meaning sweet).

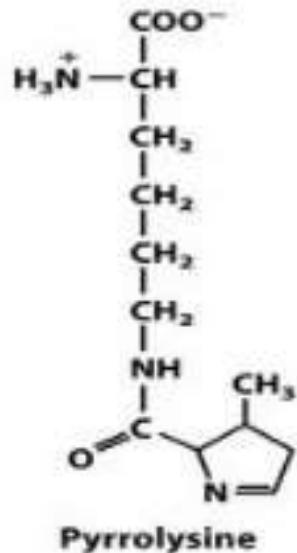
New Amino Acids

In addition to 20 L-amino acids that take part in protein synthesis, recently two more new amino acids described. They are:

A. Selenocysteine - 21st amino acid it occurs at the “active site” of several enzymes such as (*Glutathione peroxidase*).

B. Pyrrolysine - 22nd amino acid.

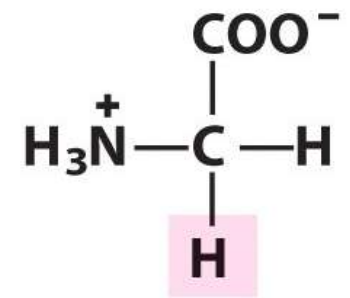
The STOP codon UAG can code for pyrrolysine. Pyrrolysine exists only in the archaeobacterium.



Optical properties of amino acids

Because the carbons of 19 of the 20 amino acids (except glycine) are attached to four different groups (i.e., a hydrogen, a carboxyl group, an amino group, and an R group), they are referred to as asymmetric, or chiral, carbons.

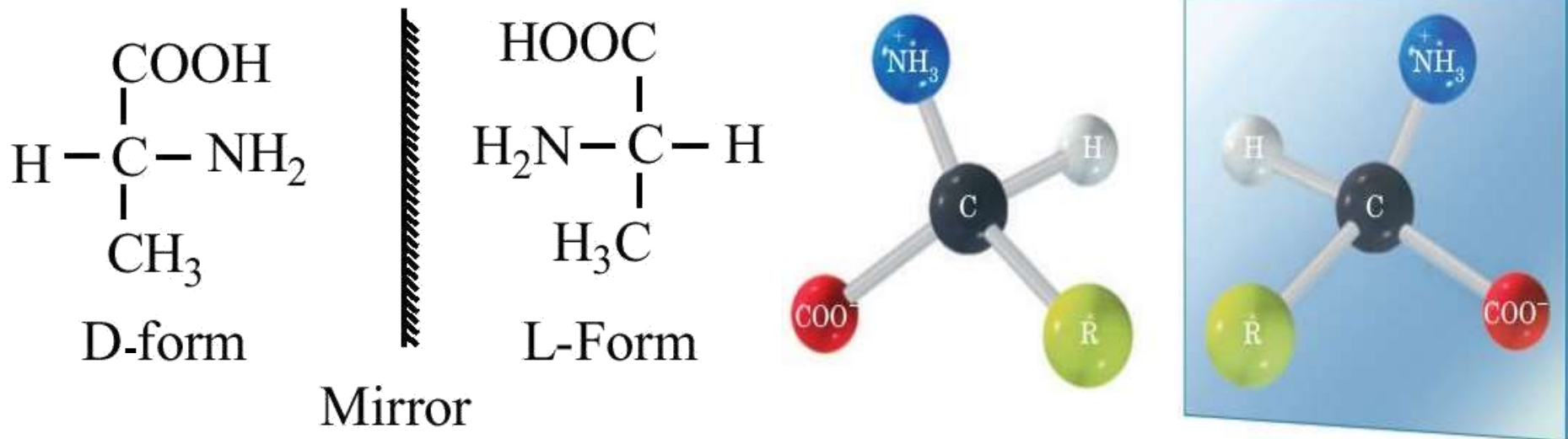
Glycine is symmetrical molecule because its carbon is attached to three different groups not four.



Glycine

Based on the position of amino group on the asymmetric carbon atom, amino acids exist in two types, they are Dextro and Levo amino acids.

The two forms in each pair are termed stereoisomers or enantiomers.



The amino acid having the NH_2 group on the right is called dextro or D-amino acid. The amino acid having the NH_2 group on the left is called levo or L-amino acid.

All proteins amino acid are alpha (α) amino acids because the NH_2 group is attached to the α carbon atom.

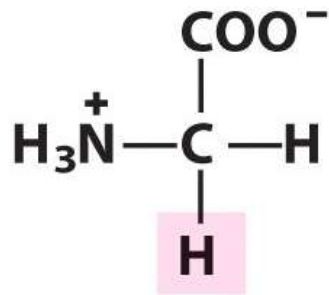
Amino acid residues in protein molecules are exclusively L-stereoisomers.

Although D-amino acids occur in nature, most often in bacterial cell walls and in some antibiotics, they are not found in proteins.

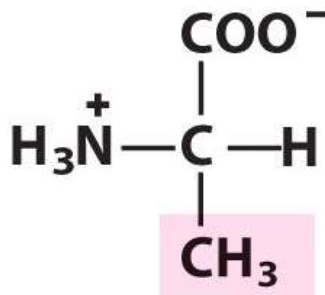
Classification of amino acids

On the basis of the composition of the side chain or R group the (20) amino acids can be classified to:-

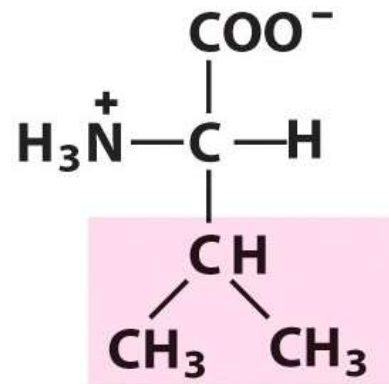
1- Aliphatic amino acids



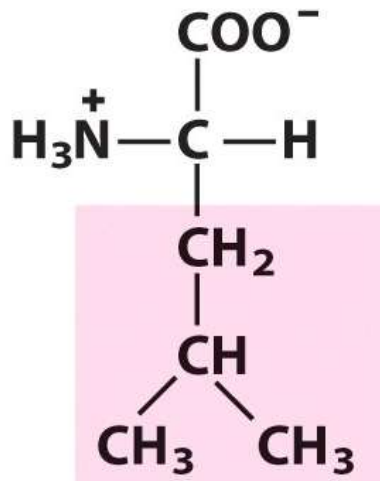
Glycine



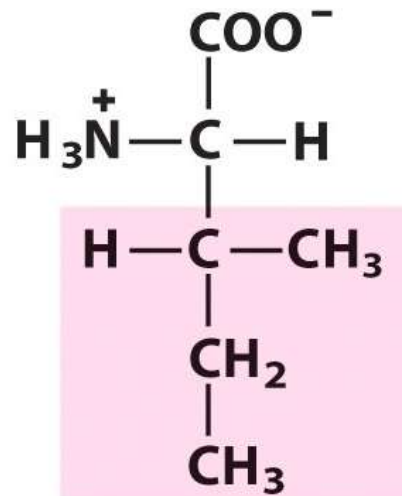
Alanine



Valine



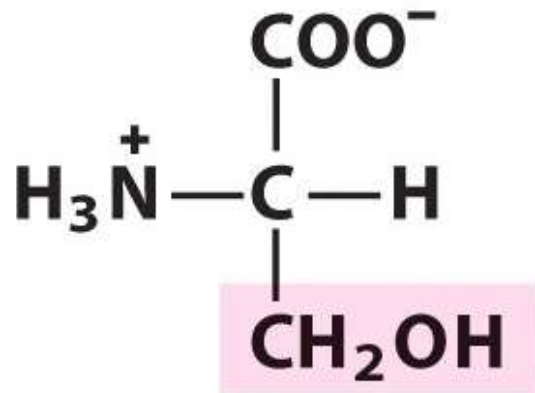
Leucine



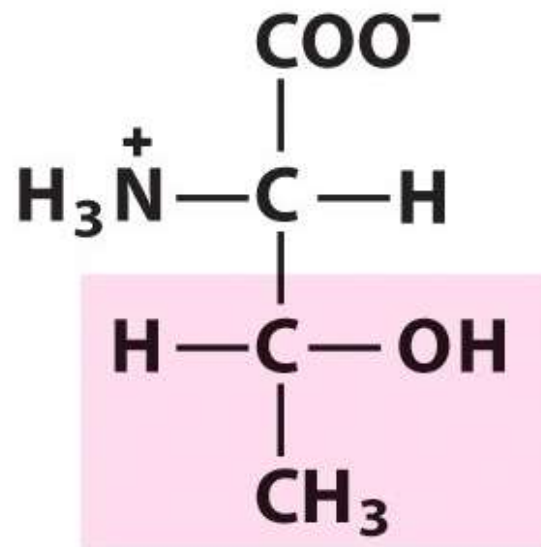
Isoleucine

2-Amino acids containing hydroxyl (-OH) group

- Serine
- Threonine



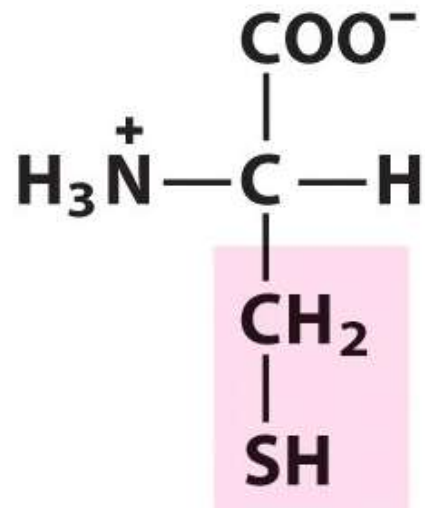
Serine



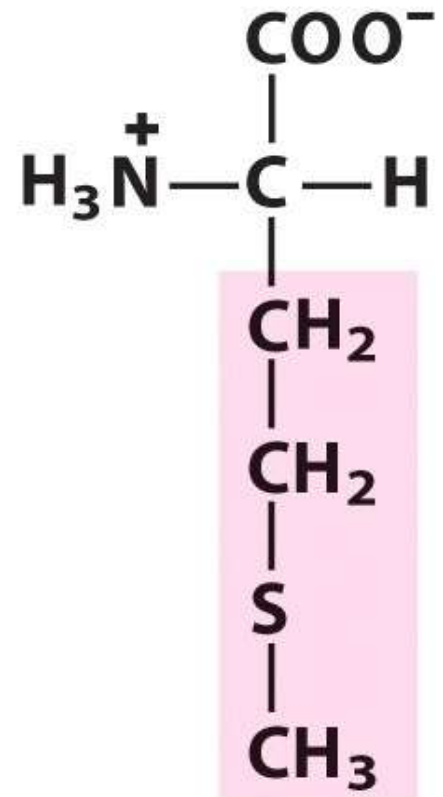
Threonine

3-Amino acids containing thiol or (-SH) group

- Cysteine
- Methionine

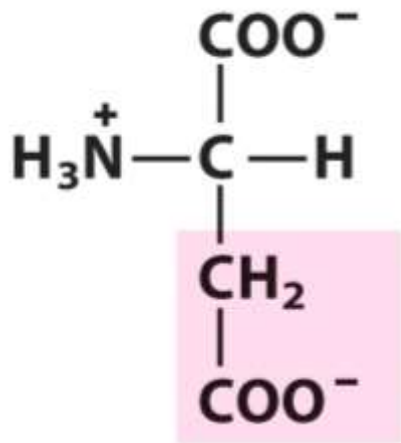


Cysteine

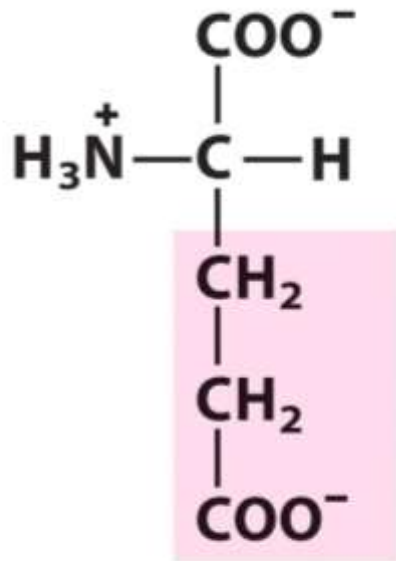


Methionine

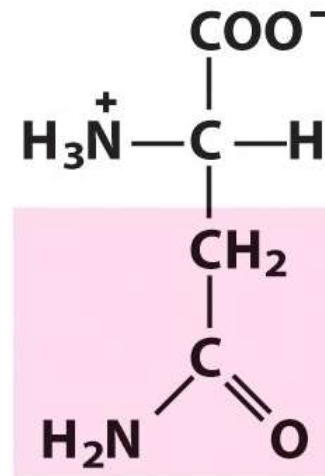
4-Acidic amino acids and its amides



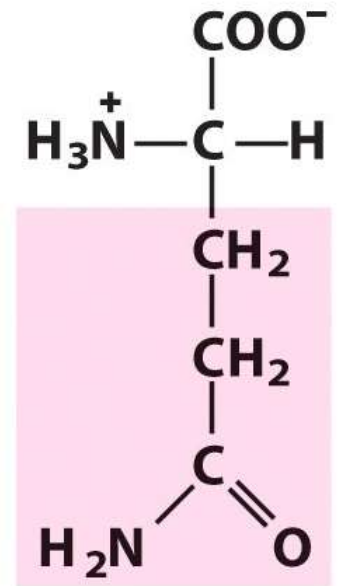
Aspartate



Glutamate

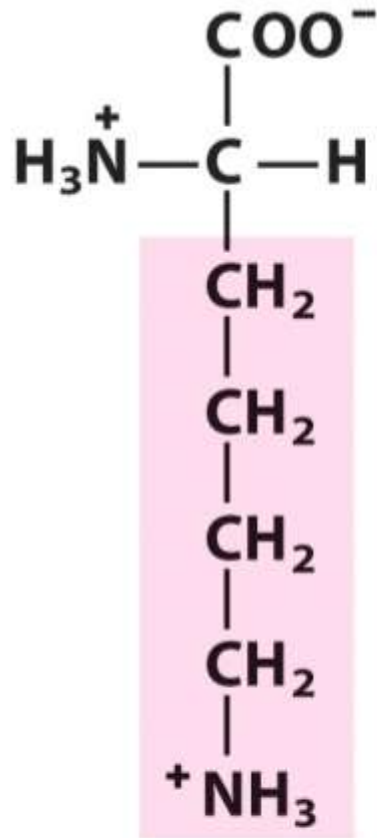


Asparagine

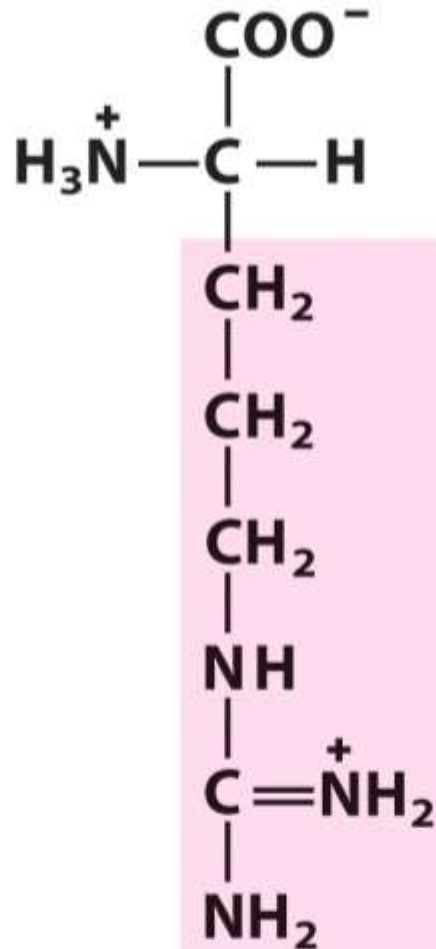


Glutamine

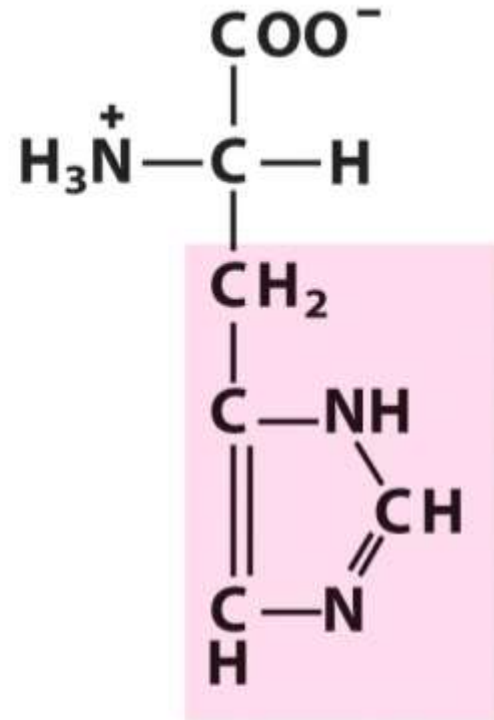
5-Basic amino acids



Lysine



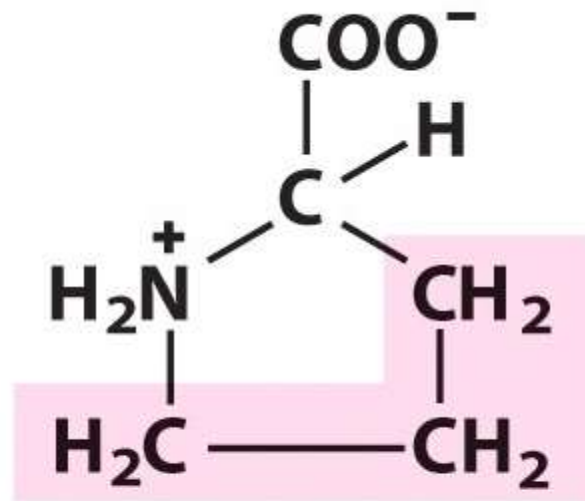
Arginine



Histidine

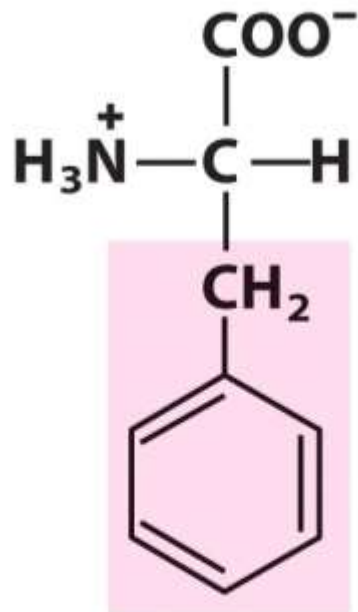
6- Heterocyclic amino acids (Containing imino '-NH-' group)

- **Proline**

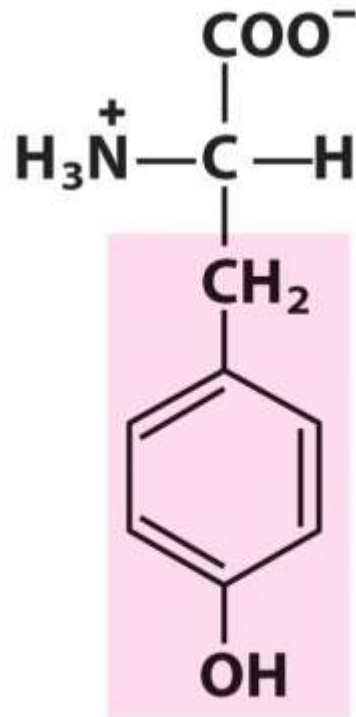


Proline

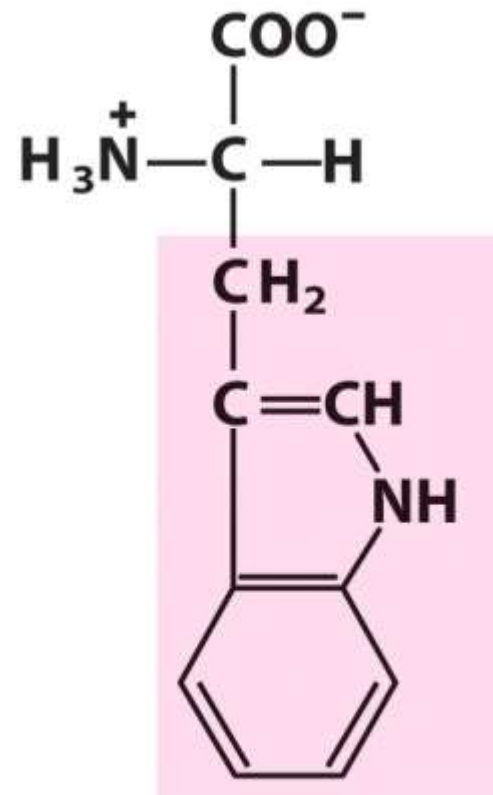
7-Aromatic amino acids



Phenylalanine



Tyrosine



Tryptophan

Classification of amino acids according to their polarity

The amino acids are classified based on the nature of their R groups, in particular their polarity or tendency to interact with water at biological pH. The polarity of the R groups varies widely, from totally non polar to highly polar.

Non-polar, aliphatic R-group

The R-group in this class of amino acids are non polar (or) hydrophobic ('water-fearing'), which are **glycine, alanine, valine, proline, leucine, isoleucine and methionine.**

Non-polar, aromatic R groups

Includes **phenylalanine, tyrosine** and **tryptophan**.

The hydroxyl group of tyrosine can form hydrogen bond with other compounds and it is an important functional group in some enzymes. Phenyl alanine is significantly more non-polar than Tyrosine and tryptophan because of the hydroxyl group of tyrosine and the nitrogen in the indole ring in tryptophan.

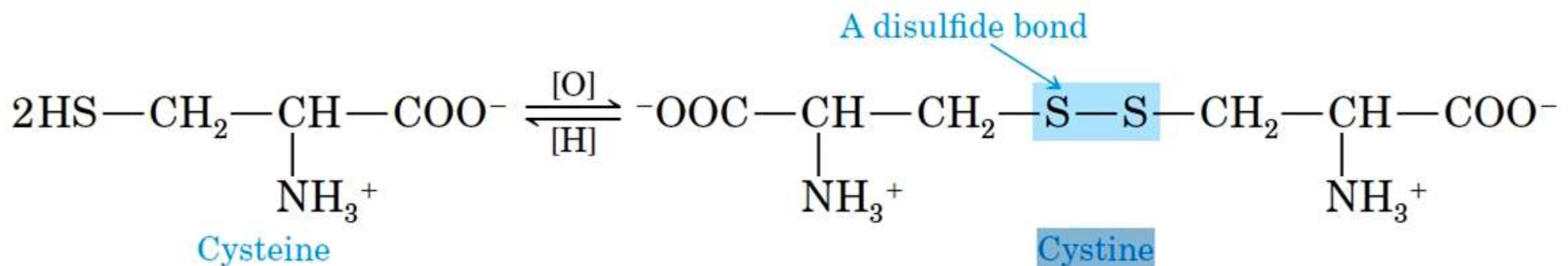
Polar-uncharged R groups

The R groups of these amino acids are more soluble in water or more hydrophilic ('water attracting'), than those of the non-polar amino acids because they contain functional groups that form hydrogen bonds with water Includes **serine, threonine, cysteine, asparagine** and **glutamine**.

The polarity of serine and threonine is contributed by their hydroxyl groups which enables them to participate in hydrogen bonding; that of cysteine by its sulphhydryl (-SH) group; and that of asparagine and glutamine by their amide groups.

Cysteine is readily oxidized to form a covalently linked dimeric amino acid called **cystine**, in which two cysteine molecules are joined by a disulfide bond.

The disulfide linkage are strongly hydrophobic (nonpolar), plays role in the structures of many proteins by forming covalent links between parts of a protein molecule or between two different polypeptide chains.



Polar- Positively charged (basic) **R-groups**

The amino acids in which the R groups have significant positive charges at pH 7.0 are **lysine, arginine** and **histidine**.

Polar-Negatively charged **(Acidic) R groups**

The two amino acids having R groups with a net negative charge at pH 7.0 are **aspartate** and **glutamate**.

Classification based on nutritional requirements in to:- essential amino acids and non-essential amino acids.

Certain amino acids can not be synthesized by our body and they must be included in the diet for normal health.

These amino acids are called essential amino acids.

For human body about 10 amino acids are considered as essential.

Certain amino acids synthesized in the cells from essential amino acids or from other compounds. They are called non-essential amino acids.

Those essential amino acids are necessary for synthesis of new body proteins during growth, continuous regeneration of tissue in adults and synthesis of hormones, enzymes, and other proteins.

The biosynthesis of proteins requires the presence of all of the protein's constituent amino acids. If any of the 20 amino acids is missing or in short supply, protein biosynthesis will be inhibited.

The body can synthesize some of the essential amino acids, but not in sufficient quantities for its needs, especially in the case of growing children (particularly children's requirement for arginine and histidine).

So Histidine and Arginine are essential for infants, small children, pregnancy and lactating women.

Classification based on nutritional requirements

A. Essential or Indispensable

The amino acids may further be classified according to their essentiality for growth.

Thus **Isoleucine, Leucine, Threonine, Lysine, Methionine, Phenylalanine, Tryptophan, and Valine** are essential amino acids. Their carbon skeleton cannot be synthesized by human beings and so preformed amino acids are to be taken in food for normal growth.

Those essential amino acids are necessary for synthesis of new body proteins during growth, continuous regeneration of tissue in adults and synthesis of hormones, enzymes, and other proteins.

B. Partially essential or Semi-essential The body can synthesize some of essential amino acids, but not in sufficient quantities for its needs, especially in the case of growing children (particularly children's requirement for **arginine and histidine**).

So **Histidine and Arginine** are essential for infants, small children, pregnancy and lactating women.

But they are not essential for the adult individual.

C. Non-essential or Dispensable

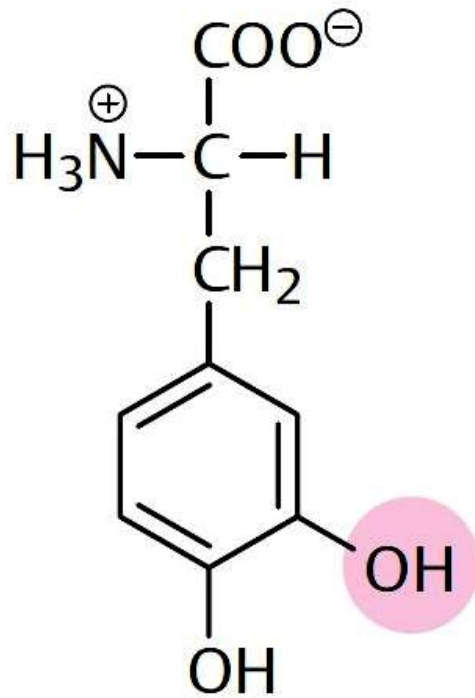
The remaining 10 amino acids are non-essential, because their carbon skeleton can be synthesized

Classification based on nutritional requirements.

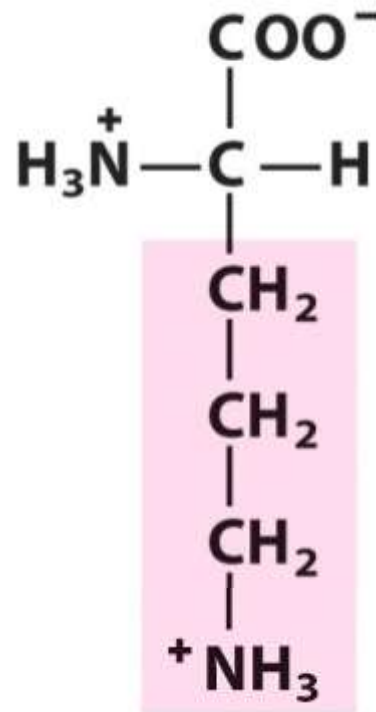
Essential	Non essential
Valine Leucine Isoleucine Threonine Methionine Lysine Phenylalanine tryptophan	Glycine Alanine Serine Cysteine Aspartic acid Asparagine Glutamic acid Glutamine Tyrosine Proline

Partially essential or Semi-essential
Histidine and Arginine

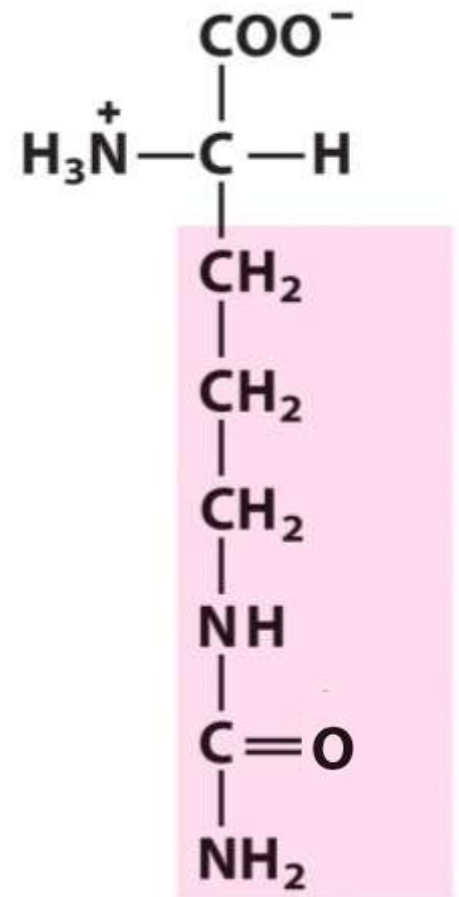
Certain amino acids which do not exist in proteins are called non protein amino acids. For example:- β -alanine



L-Dopa



Ornithine



Citrulline

PROTEIN BIOSYNTHESIS

- Proteins are linear polymer of amino acid.
- Proteins may have thousands of amino acid residues.

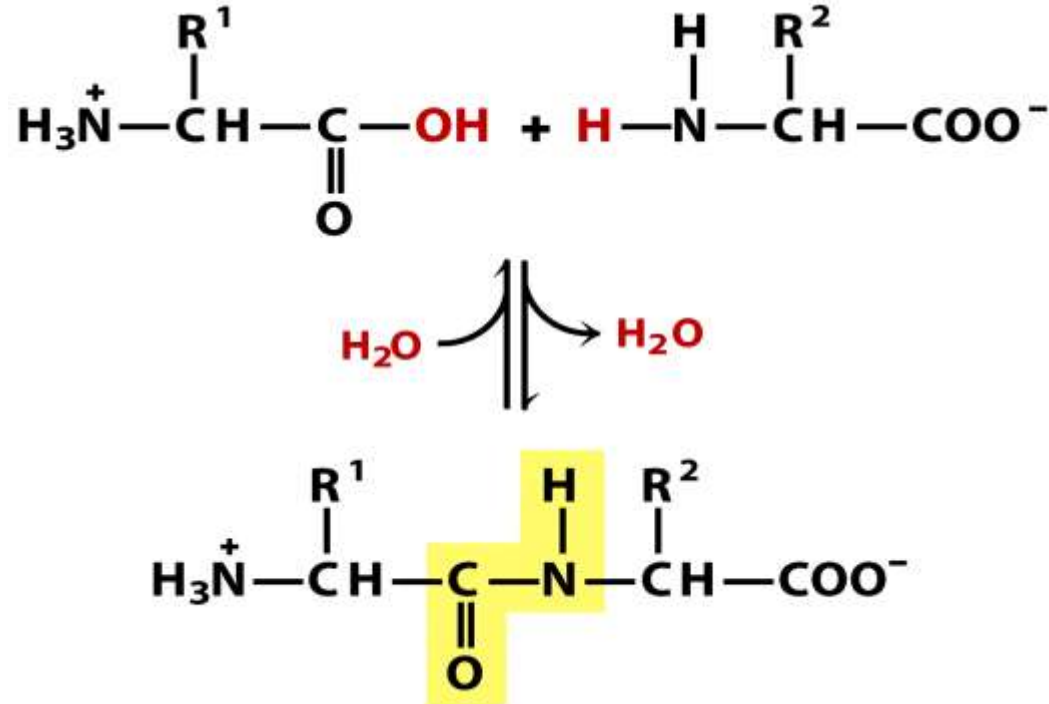
Alpha carboxyl group of one amino acid reacts with alpha amino group of another amino acid to form a peptide bond or CO-NH

Proteins are made by polymerisation of amino acids through peptide bonds.

Two amino acids are combined to form a **dipeptide**; three amino acids form a **tripeptide**.

Four will make a **tetrapeptide**; a few amino acids together will make an **oligopeptide**; and combination of **10** or more amino acids is called as a **polypeptide**.

Big polypeptide chains containing more than 50 amino acids are called proteins.



Students should have a clear concept of the term "sequence". See the following example:

Gly - Ala - Val (1)

Gly - Val - Ala (2)

Both the tripeptides shown above contain the same amino acids; but their sequence is altered. When the sequence is changed, the peptide is also different.

In a tripeptide, there are 3 amino acids, but these 3 can be any of the total 20 proteogenic amino acids.

Thus $20^3 = 8000$ different permutations and combinations are possible in a tripeptide.

Examples on Peptides:

1- Dipeptide (tow amino acids joined by one peptide bond):

Example: **Aspartame** which acts as sweetening agent being used in replacement of cane sugar. It is composed of aspartic acid and phenyl alanine.

2- Tripeptides (3 amino acids linked by two peptide bonds).

Example: **Glutathione (GSH)** which is formed from 3 amino acids: glutamic acid, cysteine and glycine. It helps in absorption of amino acids, protects against hemolysis of RBC by breaking H_2O_2 which causes cell damage.

3- Nonapeptides: (9 amino acids)

Examples: Two hormones; oxytocine and vasopressin (ADH).

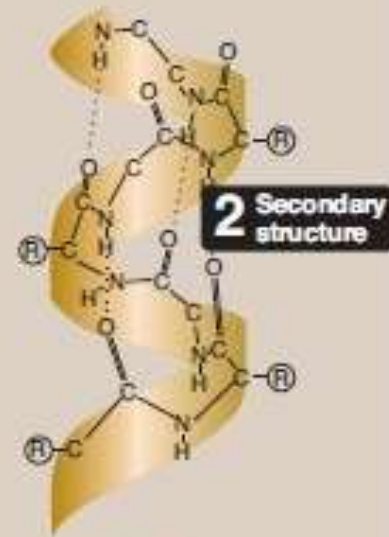
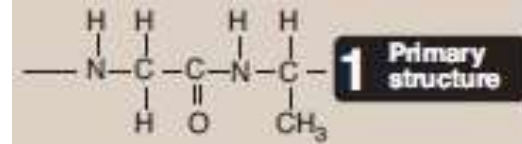
4- Polypeptides: 10 or more amino acids: e.g. Insulin hormone

Structure of proteins

Linder strom - Lang suggest four types of structural organization for proteins.

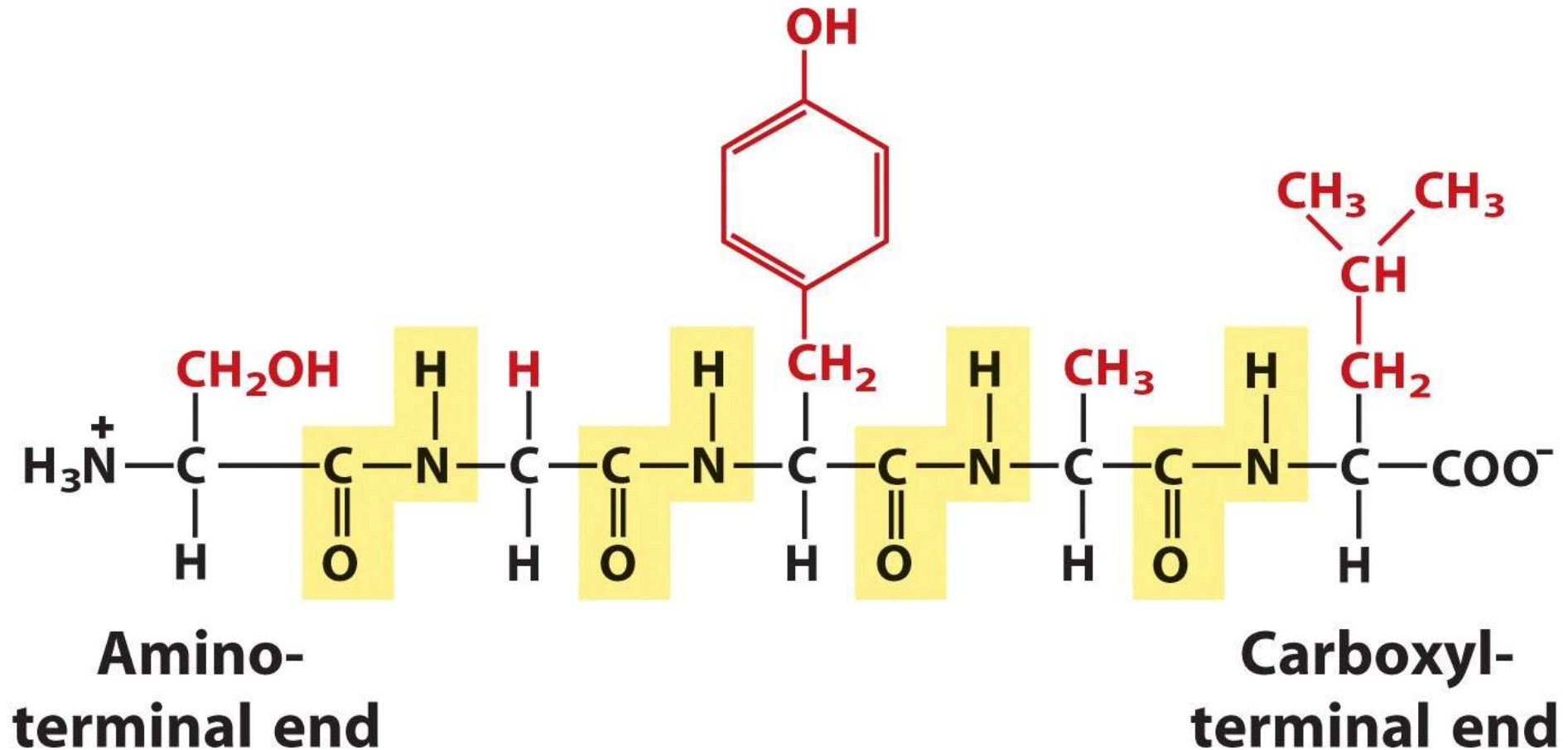
They are

1. Primary structure.
2. Secondary structure.
3. Tertiary Structure.
4. Quaternary structure.



1- Primary structure:

The primary structure of protein is defined as the sequence of amino acid residues making up its polypeptide chain.



The peptide bond is covalent in nature, quiet stable and referred as backbone of the protein.

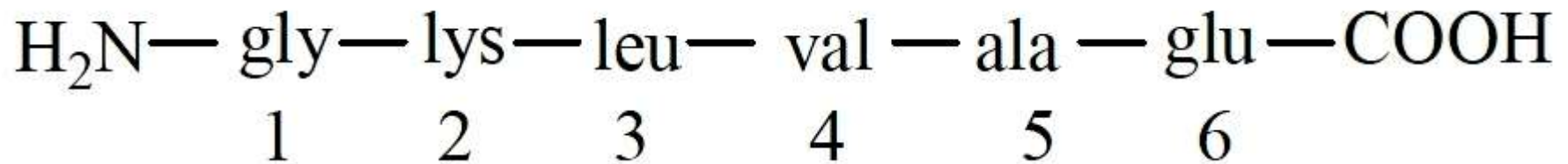
Each peptide or protein in the body has its own unique (primary structure) sequence of amino acids.

The amino acids are arranged in specific sequence in these polypeptide chain.

Primary structure of a protein determines to a large extent the native of secondary and tertiary structures.

The amino acid with free NH_3^+ group is known as amino terminal or (N-terminal) amino acid, and the amino acid with free COO^- group is known as the carboxyl or (C-terminal).

The amino acids in a polypeptide chain are numbered from the N-terminal end.

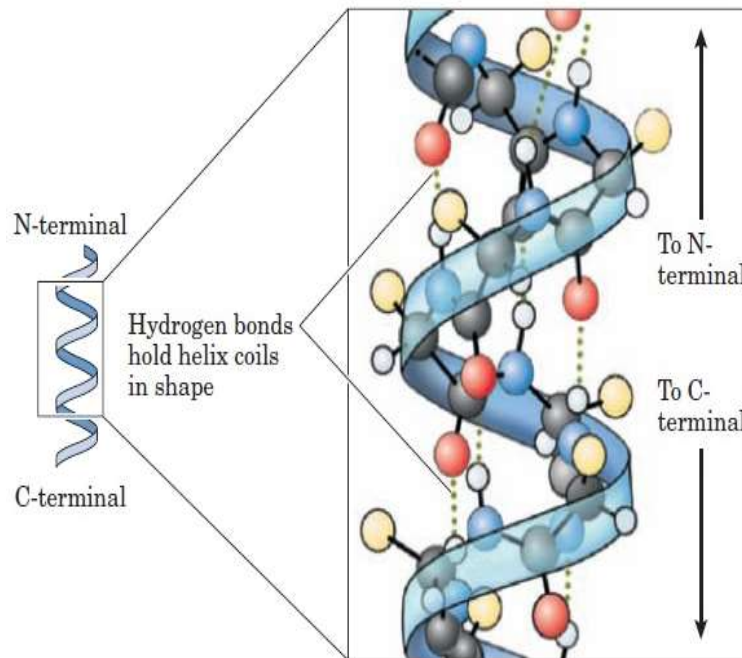


2- Secondary structure:

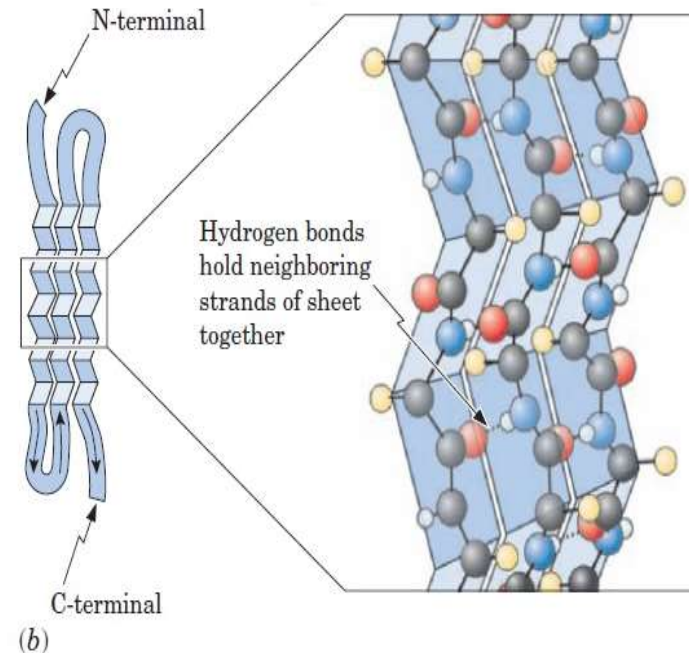
The folding of polypeptide chains into regular structure such as α -helixes and β -pleated sheets.



C = gray
N = blue
O = red
R = green



(a)



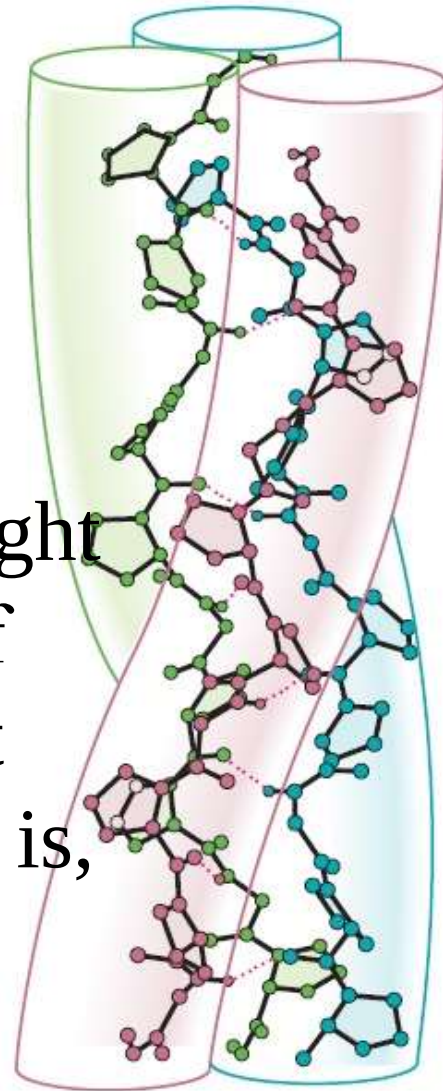
(b)

Secondary structure is the result of hydrogen bonding between the amide hydrogen's and carbonyl oxygen's of the peptide bond. Many hydrogen bonds are needed to maintain the secondary structure, the two most common types of secondary structure are the α -helixes and β -pleated sheets because they maximize the hydrogen bonding in the backbone.

Collagen is classified as a secondary structure (Triple Helix).

Collagen is the structural protein of connective tissues (bone, cartilage, tendon, blood vessels, skin), where it provides strength and elasticity.

Collagen is the most abundant protein in humans, it makes up about 30% by weight of all the body's protein. Each strand of collagen consists of repetitive units that can be symbolized as Gly—X—Y; that is, every third amino acid in the chain is glycine.



When collagen is synthesized, the amino acids proline and lysine are incorporated into the chain of amino acids. These are later modified by two enzymes to form 4-hydroxyproline and 5-hydroxylysine which account for nearly one-fourth of the amino acids in collagen.

Both of these enzymes require vitamin C to carry out these reactions. In fact, this is the major known physiological function of vitamin C. Without hydroxylation, hydrogen bonds cannot form and the triple helix is weak, resulting in fragile blood vessels

Does α -helices of secondary structure affect rigidity of protein or not?

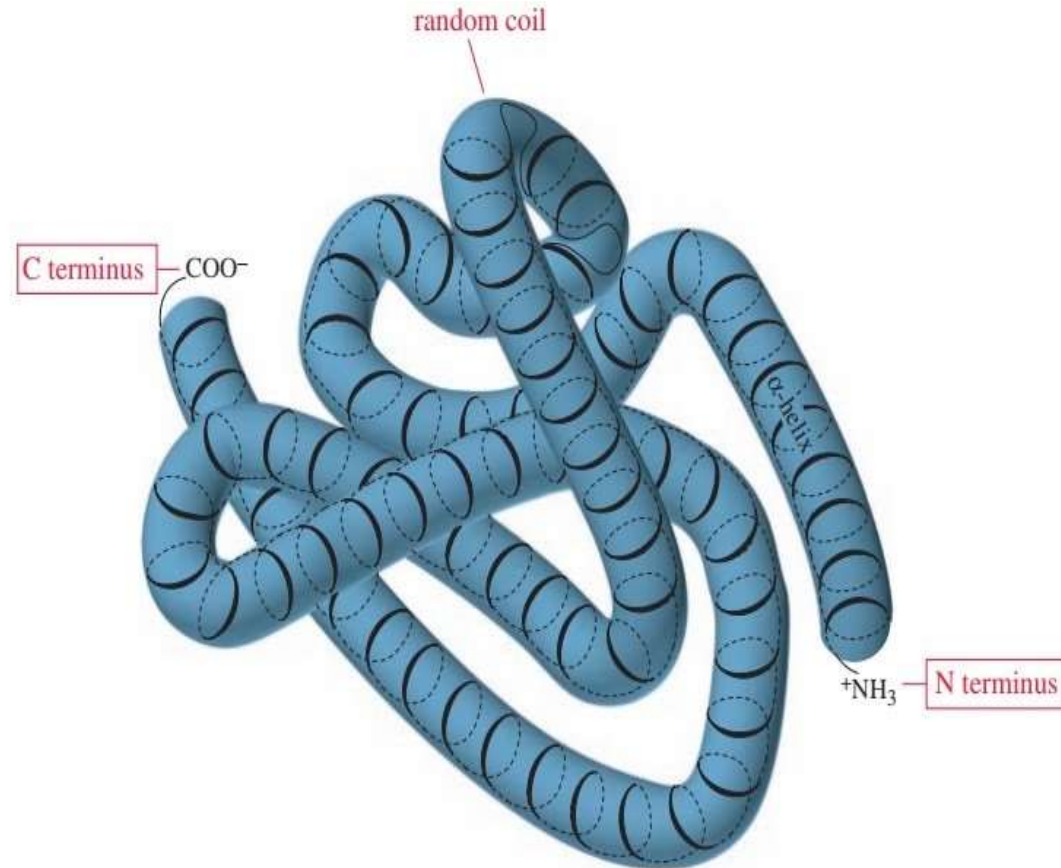
Diverse group of proteins contain α -helices. For example, the keratins are a family of fibrous proteins whose structure is nearly entirely α -helical. They are a major component of tissues such as hair and skin

In contrast to keratin, myoglobin, whose structure is also highly α -helical, is a globular, flexible molecule.

So the rigidity of protein is determined by the number of disulfide bonds between the constituent polypeptide chains.

3- Tertiary (Ternary) structure:-

The polypeptide chain with secondary structure may be further folded (super-folded) or twisted about itself forming many sizes.

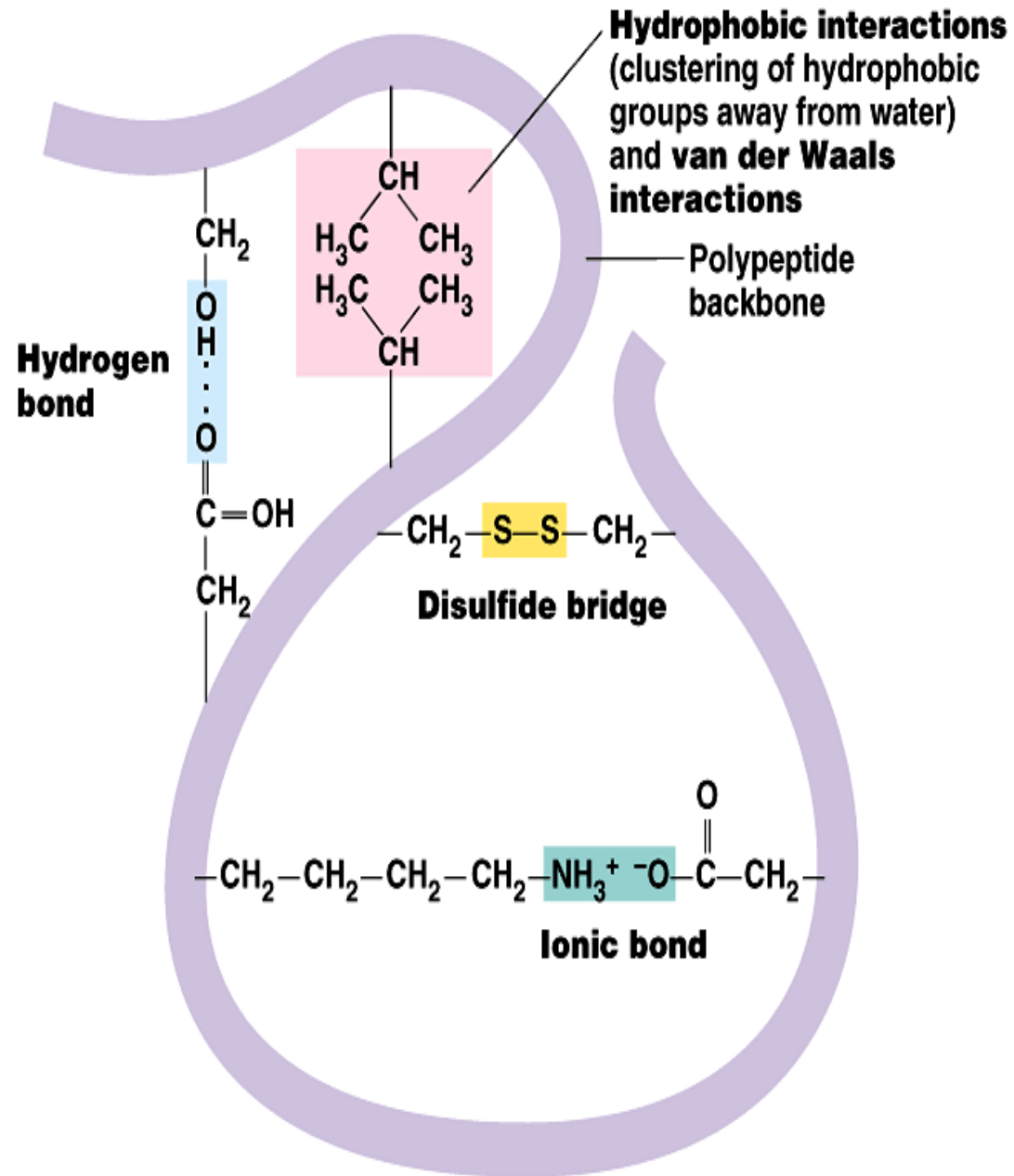


It determined by a variety of interactions (bond formation) among R groups and between R groups and the polypeptide backbone.

a. The weak interactions include:

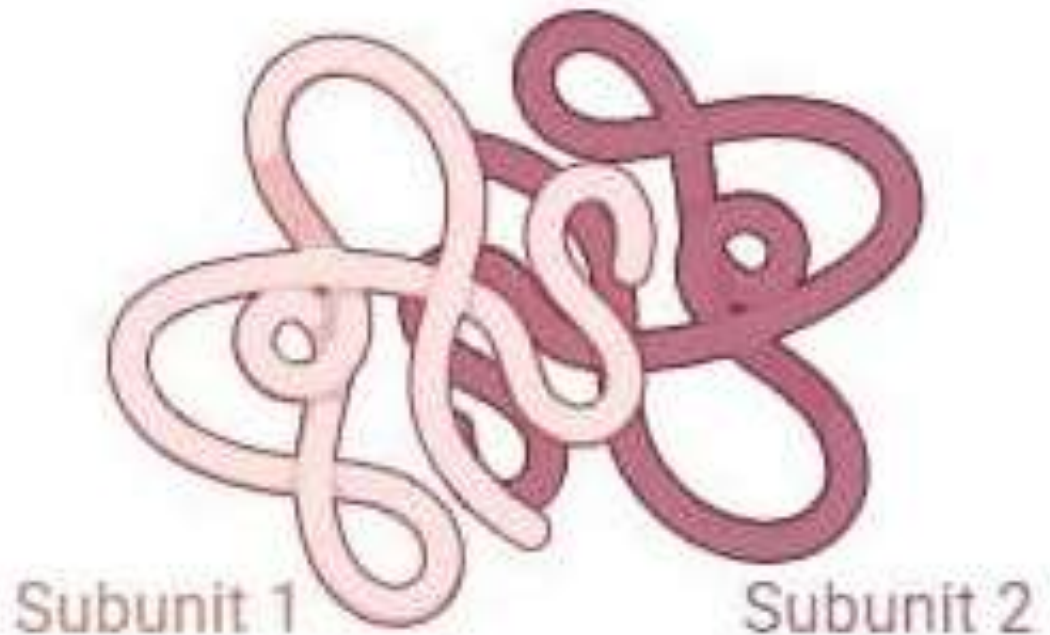
- **Hydrogen bonds** among polar side chains
- **Ionic bonds** between charged R groups (basic and acidic amino acids)
- **Hydrophobic interactions** among hydrophobic (non polar) R groups.

b. Strong covalent bonds include: disulfide bridges, that form between the sulfhydryl groups (SH) of cysteine monomers, stabilize the structure.



4) Quaternary structure:

Some proteins are made up of more than one polypeptide chain. These peptide chains held together by non-covalent interactions or by covalent links, is called quaternary structure.



Each constituent peptide chain is called as a subunit. The subunits can be identical or quite different in their primary, secondary or tertiary structure.

The forces that hold the quaternary structure are the same as those in the tertiary structure. (hydrogen bond, ionic bridge, van der waals forces and disulfide bridge).

In some cases the quaternary structure of a functional protein involves binding to a non-protein group. This additional group is called prosthetic group.

For example many of the receptor proteins on cell surfaces are glycoproteins; these are proteins with sugar groups covalently attached.

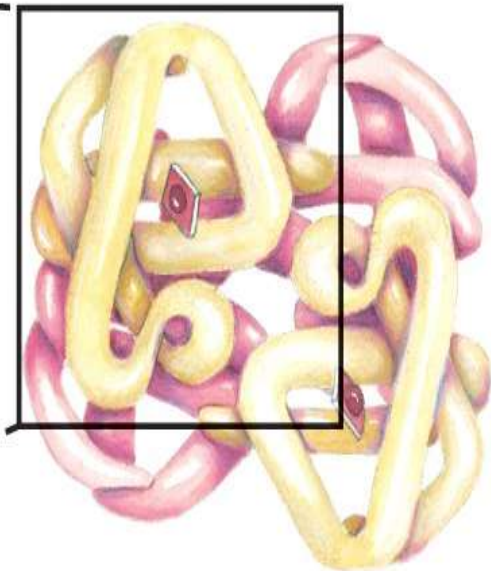
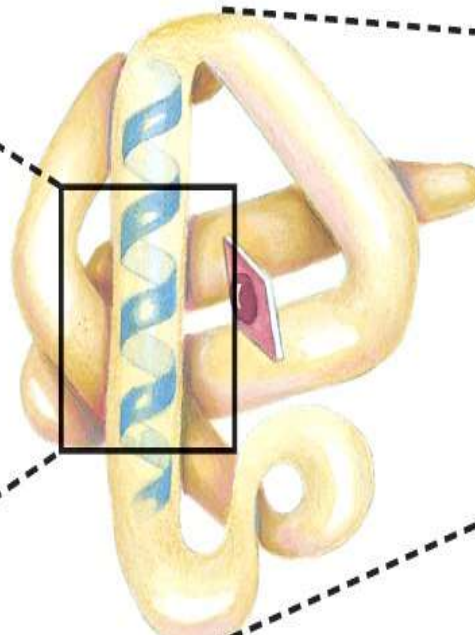
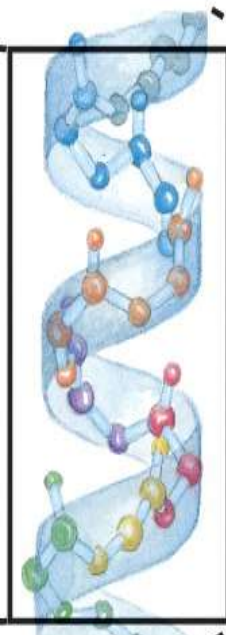
Levels of protein structure

**Primary
structure**

**Secondary
structure**

**Tertiary
structure**

**Quaternary
structure**



**Amino acid
residues**

α Helix

Polypeptide chain

Assembled subunits

Examples of multisubunit enzymes

Enzyme	Number of Subunits
HIV protease	2
Hexokinase	2
Bacterial cytochrome oxidase	3
Lactate dehydrogenase	4
Aspartate carbamoyl transferase	12
Human cytochrome oxidase	13

Types of protein:-

- 1. Complete proteins:** Protein derived from animal sources is generally complete proteins because it contain all essential and non essential amino acids in their structure. Such as meat, fish and eggs.
- 2. Incomplete proteins:** Protein derived from vegetable sources is generally incomplete proteins because it lacks a sufficient amount of one or more essential amino acids in their structure. Such as nuts, seeds and corn.
Zein corn storage protein lacks tryptophan and lysine.

CLASSIFICATION OF PROTEINS

On the basis of shape and size:

1- Fibrous proteins: When the axial ratio of length: width of a protein molecule is more than 10, it is called a fibrous protein.

Examples: α -keratin from hair and collagen.

2- Globular protein: When the axial ratio of length: width of a protein molecule is less than 10, it is called as globular protein.

Examples: Myoglobin, haemoglobin, ribonuclease, etc.

Classification of proteins according to its composition:-

I- Simple proteins:

i.e. on hydrolysis gives only amino acids

Examples:

1- Albumin and globulins: present in egg, milk and blood

They are proteins of high biological value i.e. contain all essential amino acids and easily digested.

Types of globulins: α 1 globulin: e.g. antitrypsin

α 2 globulin: e.g. hepatoglobulin: protein that binds hemoglobin to prevent its excretion by the kidney

β -globulin: e.g. transferrin: protein that transport iron

γ -globulins = **Immunoglobulins** (antibodies).

2- Globins (Histones): They are basic proteins rich in histidine amino acid.

They are present in : a - combined with DNA
 b - combined with heme to form hemoglobin of RBCs.

3- Gliadines are the proteins present in cereals.

4- Scleroproteins: They are structural proteins, not digested.
include: keratin, collagen and elastin.

a- α -keratin: protein found in hair, nails, enamel of teeth and outer layer of skin.

- It is α -helical polypeptide chain, rich in cysteine and hydrophobic (non polar) amino acids so it is water insoluble.

b- collagens: protein of connective tissues.

C- Elastin: present in walls of large blood vessels (such as aorta). It is very important in lungs, elastic ligaments, skin, cartilage, ..

It is elastic fiber that can be stretched to several times as its normal length.

Structure: composed of 4 polypeptide chains (tetramer), similar to collagen being having 33% glycine and rich in proline but in that it has low hydroxyproline and absence of hydroxylysine.

2- Conjugated proteins

i.e. On hydrolysis, give protein part and non protein part and subclassified into:

1- Phosphoproteins: These are proteins conjugated with phosphate group. Phosphorus is attached to OH group of serine or threonine.

e.g. Casein of milk and vitellin of yolk.

2- Lipoproteins:

These are proteins conjugated with lipids.

Functions:

- a- help lipids to transport in blood
- b- Enter in cell membrane structure helping lipid soluble substances to pass through cell membranes.

3- Glycoproteins:

proteins conjugated with sugar (carbohydrate)

e.g. – Mucin

- Some hormones such as erythropoietin
- present in cell membrane structure
- blood groups.

4- Nucleoproteins: These are basic proteins (e.g. histones) conjugated with nucleic acid (DNA or RNA).

- e.g.
- a- chromosomes: are proteins conjugated with DNA
 - b- Ribosomes: are proteins conjugated with RNA

5- Metalloproteins: These are proteins conjugated with metal like iron, copper, zinc,

a- Iron-containing proteins: Iron may present in heme such as in - hemoglobin (Hb)

- myoglobin (protein of skeletal muscles and cardiacmuscle),
- cytochromes,
- catalase, peroxidases (destroy H₂O₂)
- tryptophan pyrrolase (desrtroy indole ring of tryptophan).

Iron may be present in free state (not in heme) as in:

- Ferritin: Main store of iron in the body. ferritin is present in liver, spleen and bone marrow.
- Hemosidrin: another iron store.
- Transferrin: is the iron carrier protein in plasma.

b- Copper containing proteins:

- Ceruloplasmin which oxidizes ferrous ions into ferric ions.
- Oxidase enzymes such as cytochrome oxidase.

c- Zn containing proteins: e.g. Insulin and carbonic anhydrase.

d- Mg containing proteins:e.g. Kinases and phosphatases.

6-Chromoproteins: These are proteins conjugated with pigment. e.g.

- All proteins containing heme (Hb, myoglobin,)
- Melanoprotein:e.g proteins of hair or iris which contain melanin.

3- Derived proteins:- These are proteins derived by partial to complete hydrolysis from the simple or conjugated proteins by the action of acids, alkalies or enzymes.. example:- fibrin from fibrinogen ,cooked meat and cooked albumin.

Gelatin: from hydrolysis of collagen

((When collagen is heated with water or dil. HCl it will be converted into gelatin which is soluble , digestible and used as diet (as jelly)).

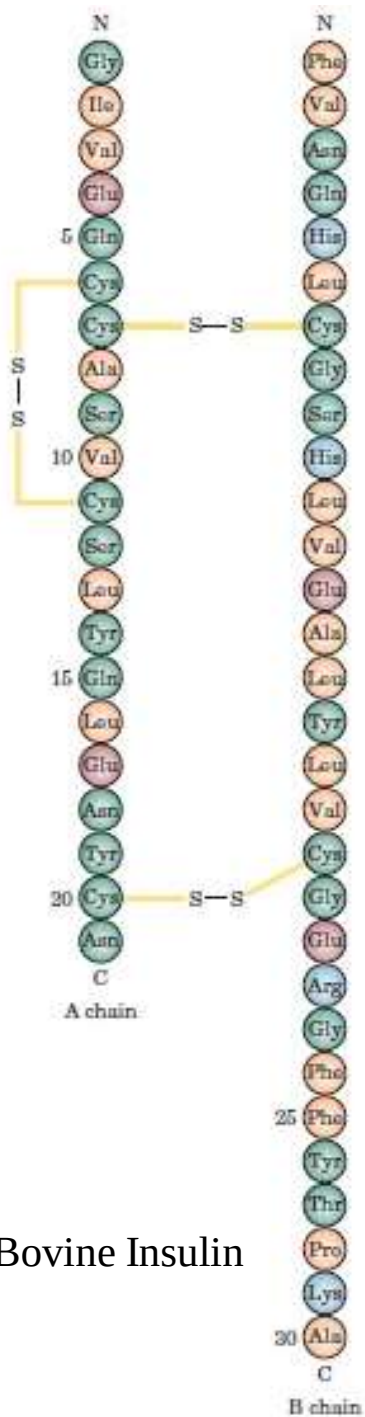
Can a protein or peptide perform the same function with some difference in its sequence?

An example is the hormone insulin.

Human insulin consists of two chains having a total of 51 amino acids. The two chains are connected by disulfide bonds. Insulin is necessary for proper utilization of carbohydrates, and people with severe diabetes must take insulin injections. So bovine insulin or insulin from hogs or sheep is used instead of human insulin.

Insulin from these sources is similar, but not identical to human insulin. The differences are entirely in the **8, 9, and 10** positions of the A chain and the C-terminal position (30) of the B chain. Despite the slight differences in structure, all of these insulin's perform the same function and even can be used by humans. However, none of the other three is quite as effective in humans as human insulin. This is one of the reasons that recombinant DNA techniques are now used to produce human insulin from bacteria.

Amino Acid Sequence Differences for Human, Bovine, Hog, and Sheep Insulin



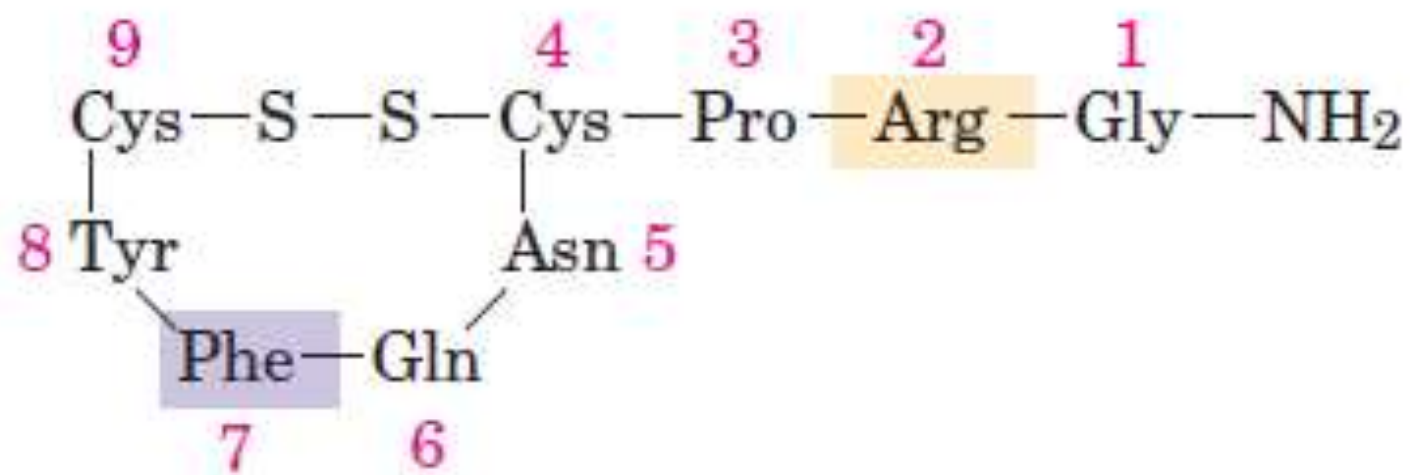
	A Chain			B Chain
	8	9	10	30
Human	—Thr—	Ser—	Ile—	—Thr
Bovine	—Ala—	Ser—	Val—	—Ala
Hog	—Thr—	Ser—	Ile—	—Ala
Sheep	—Ala—	Gly—	Val—	—Ala

Bovine Insulin

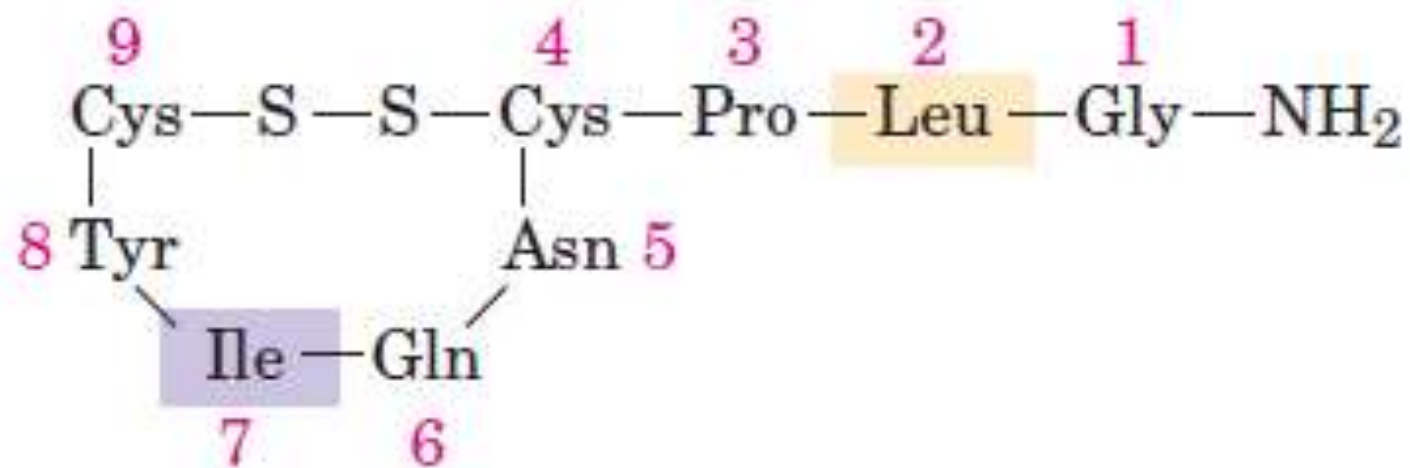
In contrast to the previous examples, some small changes in amino acid sequence make a great deal of difference. Consider two peptide hormones, oxytocin and vasopressin.

Each of these peptides contains nine amino acid residues

The difference between these two peptides is that oxytocin has a leucine residue at position 2 and an isoleucine residue at position 7, and vasopressin has an arginine residue at position 2 and a phenylalanine residue at position 7.



Vasopressin



Oxytocin

Both of these peptides have considerable physiological importance as hormones.

Oxytocin it stimulates smooth muscle contraction in the uterus during labor and in the mammary glands during lactation.

Vasopressin plays a role in the control of blood pressure by regulating contraction of smooth muscle and stimulates re-absorption of water by the kidney, thus having an antidiuretic effect. More water is retained, and the blood pressure increases.

Another instance where a minor (small) change makes a major difference is in the blood protein hemoglobin which is sickle cell anemia.

Sickle cell anemia:

Sickle cell anemia was the first human disease that was shown to be caused by mutation of a specific protein.

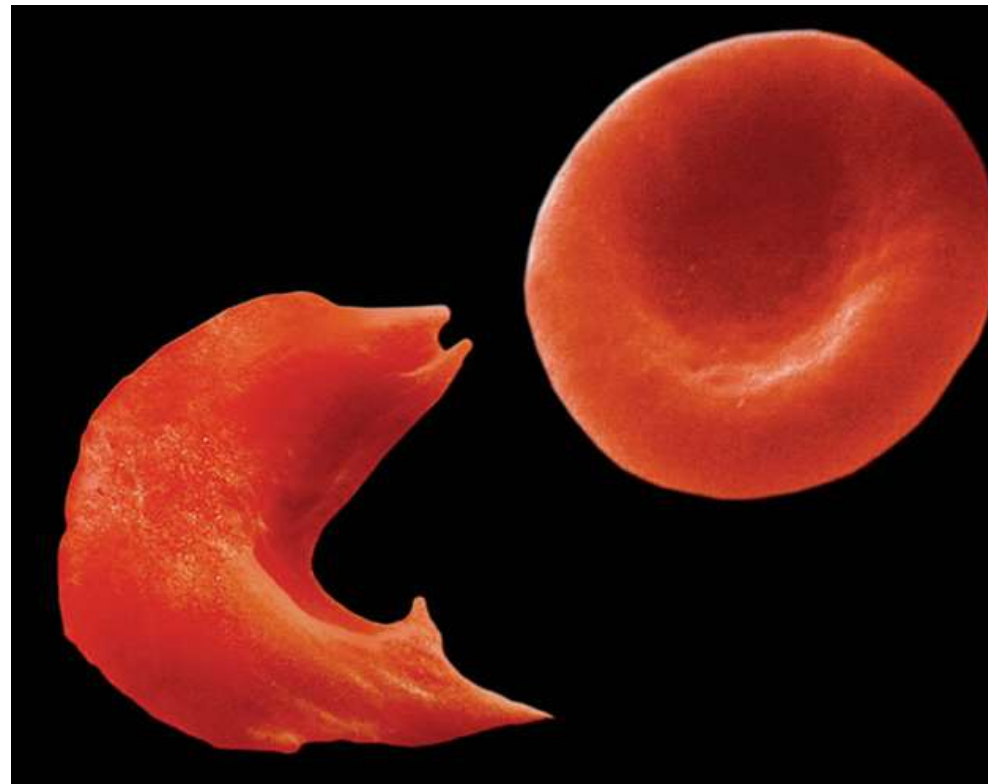
Patients with sickle cell anemia have a point mutation in their hemoglobin molecules.

Together, the four polypeptides of a hemoglobin molecule contain 574 amino acids.

The β -chains (146 AA) of normal hemoglobin have a glutamic acid residue at position 6.

In the hemoglobin from sickle cell anemia patients, this glutamic acid (negatively charged amino acid) is replaced by a valine (hydrophobic amino acid), a nonpolar amino acid.

Thus with only two amino acid changes out of 574 (one residue per β -chain), the entire biological activity of the protein is severely altered.



Sickle cell anemia receives its name from the sickled appearance of the red blood cells that form in this conditions.

The sickled cells are unable to pass through small capillaries of the circulatory system, this results in damage of many organs.

This malformation occurs in the capillaries.

As a result of this change in shape, the cells may clog (block out) the capillaries. The body's defenses destroy the clogging cells, and loss of the blood cells causes anemia.

The U.S. (FDA) Food and Drug Administration approved hydroxyurea as treatment.

Hydroxyurea prompts the bone marrow to manufacture fetal hemoglobin (HbF), which does not have beta chains where the mutation occurs. Thus red blood cells containing HbF do not sickle and do not clog the capillaries. With hydroxyurea therapy, the bone marrow still manufactures mutated sickled HbS, but the presence of cells with fetal hemoglobin dilutes the concentration of the sickling cells, thereby relieving the symptoms of the disease.

Hemoglobin is formed of 4 polypeptide chains, of which two α -chains and two β -chains, while fetal hemoglobin is composed of two α subunits and two γ (*gamma*) subunits.

Denaturation of protein:-

Denaturation is randomizing the conformation of a polypeptide chain without affecting its primary structure (peptide bonds). Or is a process in which proteins lose their structure, it occurs when there is a disruption of any of the bonds that stabilize secondary, tertiary or quaternary structure by some of external compounds,

such as treatment of proteins with strong acids or bases, high concentrations of inorganic salts, organic solvents (e.g., alcohol, acetone or chloroform), or heat.

Most proteins lose their biological function when denatured. For example, enzymes lose their activity when denatured.

Solutions such as 6 M aqueous urea, ($\text{H}_2\text{N}-\text{CO}-\text{NH}_2$), break hydrogen bonds and cause the unfolding of globular proteins.

Other chemical agents such as alcohol also denature proteins, coagulating them. This process is used in sterilizing the skin. At a concentration of 70%, ethanol penetrates bacteria and kills them by coagulating their proteins, whereas 95% alcohol denatures only surface proteins.

In living cells, some denaturation caused by heat can be reversed by chaperones. These proteins help a partially heat-denatured protein to regain its native secondary, tertiary, and or quaternary structures while others are irreversible.

Treatment of straight and curly hair:

Reducing agents, such as 2-mercaptoethanol ($\text{HOCH}_2\text{CH}_2\text{SH}$), can break the (-S-S-) disulfide bonds, reducing them to -SH groups.

The protein keratin, which makes up human hair, contains a high percentage of disulfide bonds. These bonds are primarily responsible for the shape of the hair, whether straight or curly.

In either permanent waving or straightening, the hair is first treated with a reducing agent that cleaves some of the (-S-S-) bonds. This treatment allows the molecules to lose their rigid orientations and become more flexible. The hair is then set into the desired shape, and an oxidizing agent is applied. The oxidizing agent reverses the preceding reaction, forming new disulfide bonds, which now hold the molecules together in the desired positions.

Proteins precipitation and antidotes:

Proteins can be precipitated from solutions by a variety of +ve and –ve ions. Such precipitation is of importance in the isolation of protein, in the deproteinisation of blood and other biological fluids and extracts for analysis.

An example of proteins precipitation:-

The use of AgNO_3 in cauteries is based on precipitation of proteins of tissues as Ag-salts.

Another application of this property is the **use of proteins as antidotes to metallic poisons.**

If salts of heavy metals are added, positively charged metal ions can complex with protein molecules and metal proteinates are precipitated.

Salts of Copper, Zinc, Lead, Cadmium and Mercury are toxic, because they tend to precipitate normal proteins of the gastrointestinal wall. Based on this principle, **raw egg** is sometimes used as an antidote for mercury poisoning

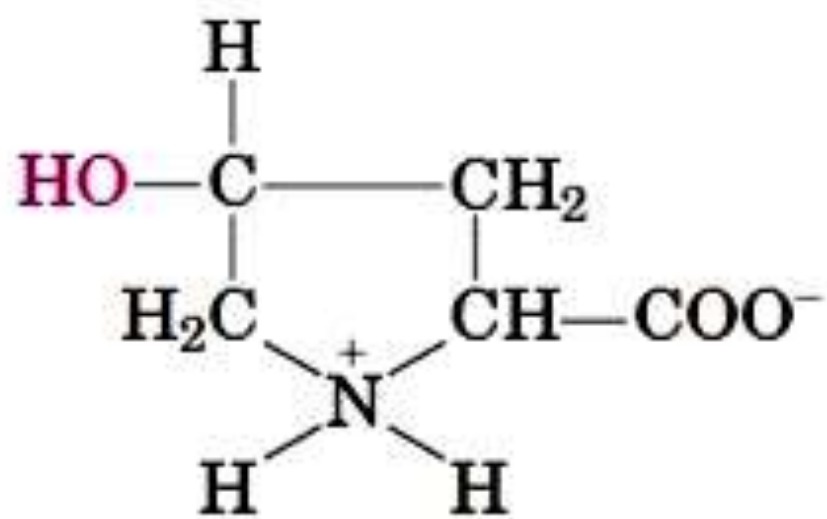
Uncommon Amino Acids in Protein:

In addition to the 20 common amino acids, proteins may contain residues created by modification of common residues already incorporated into a polypeptide.

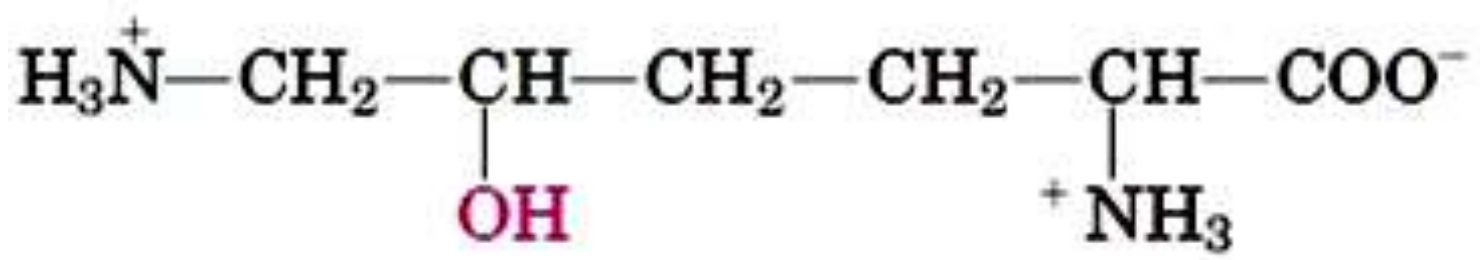
Among these uncommon amino acids are **4-hydroxyproline**, a derivative of proline, and **5-hydroxylysine**, derived from lysine.

The former is found in plant cell wall proteins, and both are found in collagen.

In many proteins some of the amino acids are modified after their incorporation; e.g., in collagen a hydroxyl group is added to each of several proline residues to yield hydroxyproline, and this does not constitute direct genetic encoding.

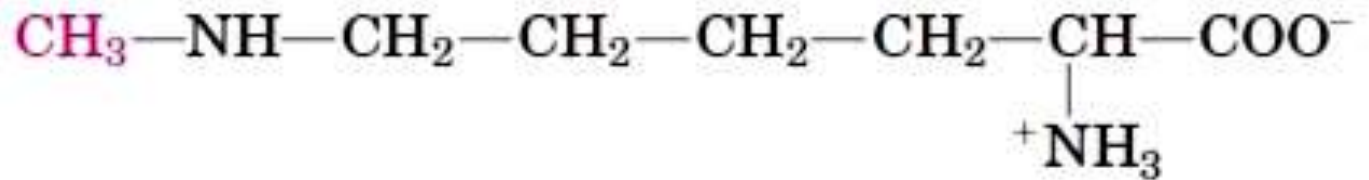


4-Hydroxyproline



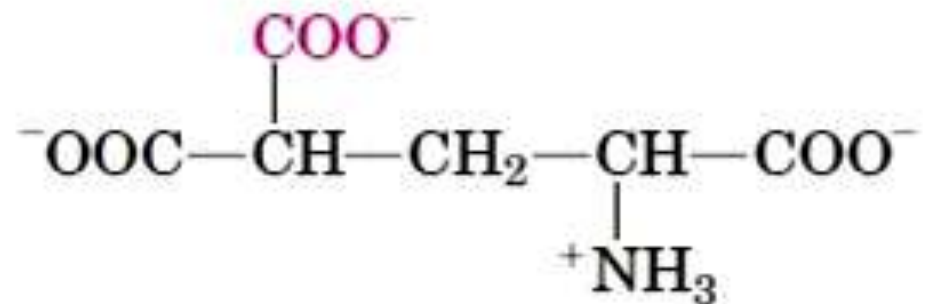
5-Hydroxylysine

6-N-Methyllysine is a constituent of myosin, a contractile protein of muscle.



6-N-Methyllysine

Another important uncommon amino acid is **γ-carboxyglutamate**, found in the blood clotting protein prothrombin.



γ-Carboxyglutamate

This modification is found, for example, in clotting factors and other proteins of the coagulation cascade.

This modification introduces an affinity for calcium ions.

In the blood coagulation cascade, vitamin K is required to γ -carboxylation of clotting factors II, VII, IX, X and protein Z.

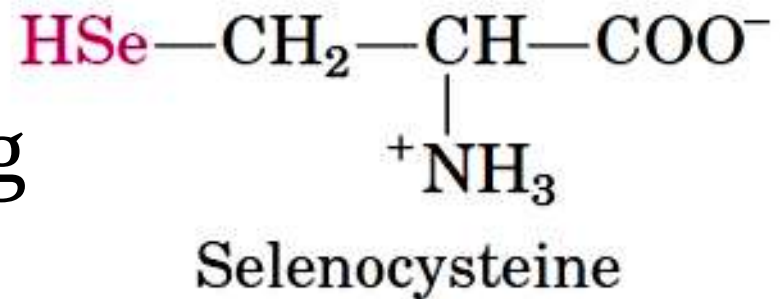
(K from koagulation, in Danish “(coagulation)”)

Selenocysteine

glutathione peroxidase that exists with high activity in erythrocytes. It contains the amino acid in its active site.

It contains selenium which replaced by the sulfur of cysteine.

Selenocysteine is not the product of a posttranslational modification, but is inserted directly into a growing polypeptide during translation.



Biologically important peptides or proteins:

1. Glutathione is a tripeptide containing glutamic acid, cysteine and glycine. It is present in erythrocytes and several other tissues. It acts as a coenzyme and protects haemoglobin against oxidation.

Glutathione is an antioxidant in plants and animals is capable of preventing damage to important cellular components caused by reactive oxygen species such as free radicals, peroxides, and heavy metals.

2. Insulin and glucagon are pancreatic hormones, involved in the regulation of glucose metabolism.
3. Albumin combine with substances of low solubilities such as cholesterol, triacylglycerol to form more soluble complexes, which can be transported in an aqueous environment of the body fluids. (it contains 584 amino acids).

4. The enzyme Renin is released from kidneys and acts on Angiotensinogen, of plasma to release a peptide **Angiotensin I** (10 amino acids) which has only a slight effect on blood pressure.

Angiotensin I is then converted to **Angiotensin II** by splitting off two amino acids which has 8 amino acid, has greater effect on blood pressure. Removal of one residue (aspartic acid) from **Angiotensin II** results in formation of **Angiotensin III** (7 amino acids) which plays role in hypertension.

Angiotensin is a peptide hormone that causes vasoconstriction and an increase in blood pressure. It is part of the renin–angiotensin system, which regulates blood pressure.

Angiotensin also stimulates the release of aldosterone from the adrenal cortex to promote sodium retention by the kidneys.

Biologically Active Amino Acids

Proteinic amino acids in addition to their primary function as components of protein, amino acids have several other biological roles.

Several -amino acids or their derivatives act as chemical messengers .

For example **gama amino butyric acid** (GABA, a derivative of glutamate, an inhibitory neurotransmitter, its underproduction is associated with epilepsy). **Serotonin**(happiness, neurotransmitter) and **melatonin** is a hormone that is produced by the pineal gland in animals and regulates sleep and wakefulness) are derivatives of tryptophan.

Thyroxine (a tyrosine derivative) and **indole acetate** (plant growth hormone) (a tryptophan derivative) are hormones chemical signal molecules produced in one cell that regulate the function of other cells.

Histidine undergoes decarboxylation to **histamine**, a powerful vasodilator in animal tissues. Histamine is released in large amounts as part of the allergic response, and it also stimulates acid secretion in the stomach.

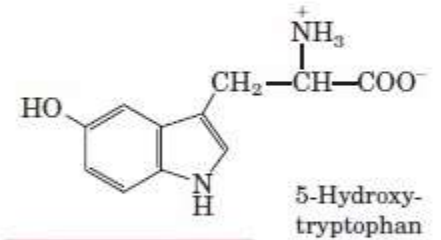
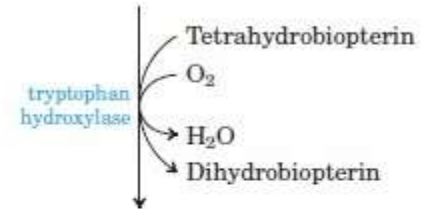
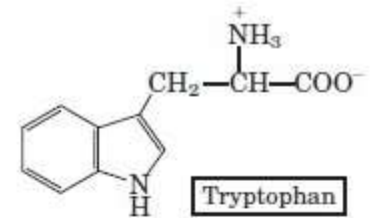
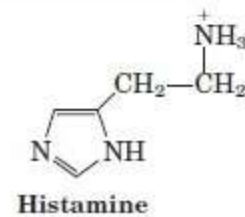
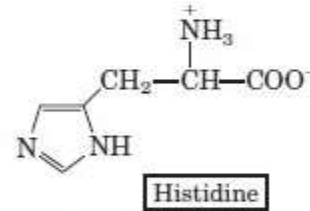
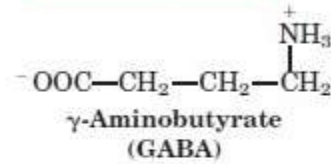
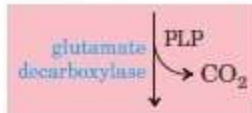
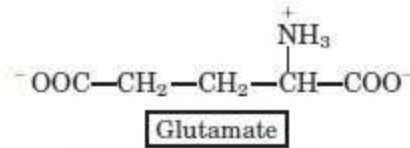
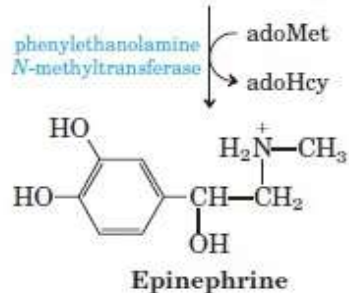
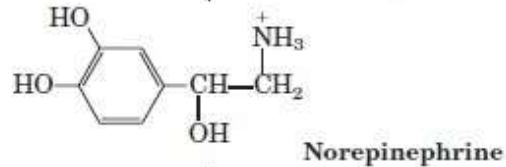
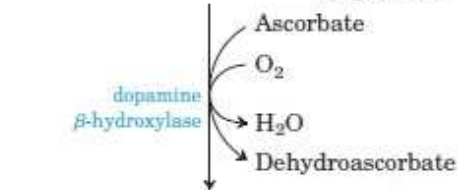
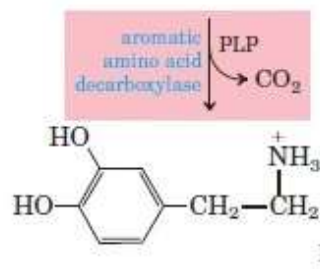
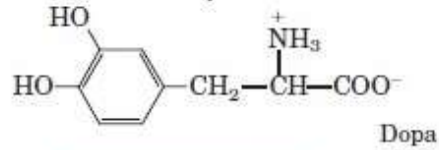
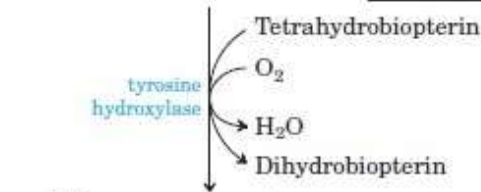
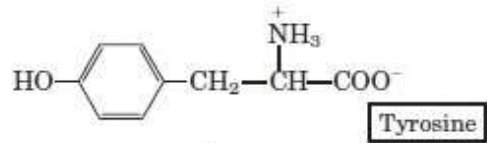
Tyrosine is a precursor to family of catecholamines that includes dopamine, norepinephrine, and epinephrine. Levels of catecholamines are correlated with changes in blood pressure.

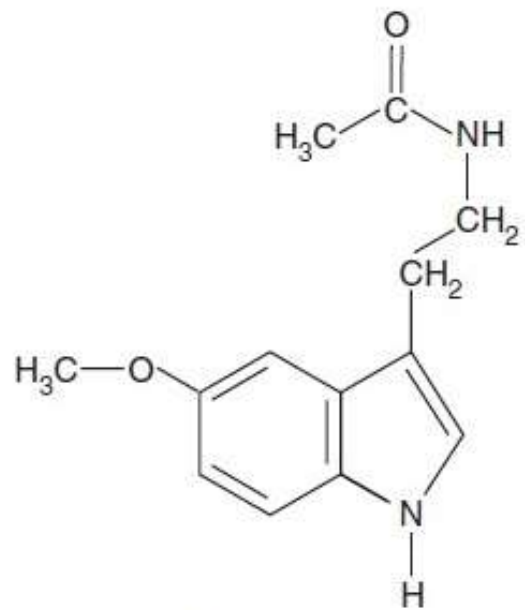
The neurological disorder Parkinson's disease is associated with an underproduction of dopamine, and it has traditionally been treated by administering L-dopa.

Cycloserine, a derivative of serine is an antituberculous drug.

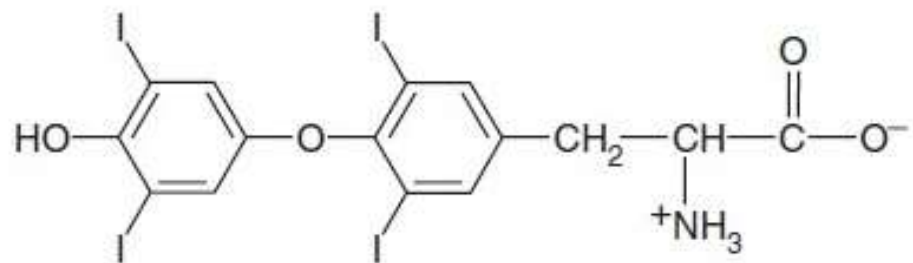
Amino acids are precursors of a variety of complex nitrogen-containing molecules.

Examples include nucleotides and the nucleic acids, heme, and chlorophyll.

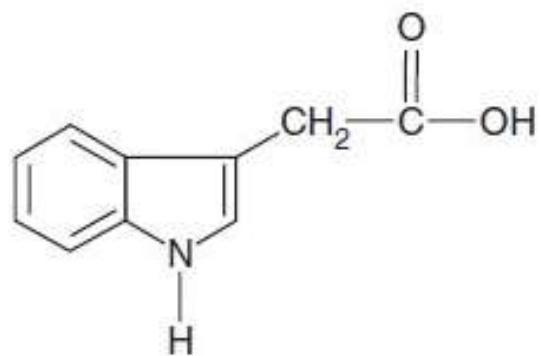




Melatonin



Thyroxine



Indole acetic acid

Aspartame, the Sweet Peptide

The dipeptide L-aspartyl-L-phenylalanine is of considerable commercial importance.

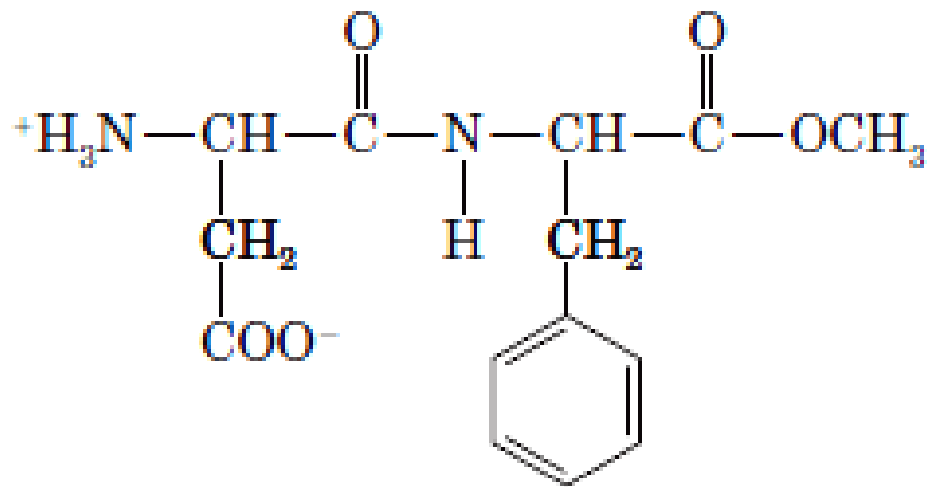
The soft-drink industry is one of the largest markets for aspartame. Aspartame is 100 to 150 times sweeter than sucrose. It is made from natural amino acids, so both the aspartic acid and the phenylalanine have the L configuration. Other possibilities have also been synthesized: the L-D, D-L and D-D configurations. However, they are all bitter rather than sweet.

You may have noticed that foods containing aspartame carry a warning about the phenylalanine portion of that artificial sweetener.

In phenylketonuria patients lack sufficient amounts of the enzyme phenylalanine hydroxylase, which catalyzes the transformation of phenylalanine to tyrosine.

As a result, large concentrations of phenylalanine, phenylpyruvate, and phenyllactate accumulate in the plasma and urine. Phenylketonuria occurs clinically in the first few weeks after birth, and if the infant is not placed on a special diet, severe mental retardation will occur.

A substitute for aspartame, which carries the trade name Alatame, contains alanine rather than phenylalanine. It has been introduced to retain the benefits of aspartame without the dangers associated with phenylalanine.



Aspartame

Amino Acid	Three-Letter Abbreviation
Alanine	Ala
Arginine	Arg
Asparagine	Asn
Aspartic acid	Asp
Cysteine	Cys
Glutamic acid	Glu
Glutamine	Gln
Glycine	Gly
Histidine	His
Isoleucine	Ile
Leucine	Leu
Lysine	Lys
Methionine	Met
Phenylalanine	Phe
Proline	Pro
Serine	Ser
Threonine	Thr
Tryptophan	Trp
Tyrosine	Tyr
Valine	Val